MILITARY SPECIFICATION

PROJECTILE, 155MM, HE, RA, M549A1 AFT (M37 EXTRUDED) PROPELLANT GRAIN FOR

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts, assembly and packing of the rocket motor aft propellant grain for the 155MM Rocket Assisted Projectile.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-T-60	- Tape, Packing, Water Proof
PPP-T-76	- Tape, Pressure Sensitive Adhesive Paper, Water Resistant (for Carton Sealing)
PPP-B-636	- Box, Shipping, Fiberboard

MILITARY

MIL-B-117	- Bag, Sleeve and Tubing - Interior Packaging
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

FSC 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-50578	-M549 Projectile, Loading, Assembling and Packing
MIL-P-63201	-Propellant, Solventless, Type M37 for Projectile, 155MM, HE, Rocket, M549
MIL-B-2427	-Box, Ammunition, Packing, Wood, Nailed
MIL-F-50449	-Filler, Sheet Form

STANDARDS

MILITARY

MIL-STD-453	-Inspection, Radiographic
MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.5)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

9324786	-Grain, Assy, Aft
9324782-2	-Propellant, Billet
8796522	-Marking Diagram and Sealing for Wood Packing Boxes

CODE OF FEDERAL REGULATIONS Title 49, Transportation Parts 100-199.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Use of Class 1 propellant. The Carpet Roll shall conform to the requirements of Military Specification MIL-P-63201.

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3.1.2 Use of Class 2 propellant. The Billet shall conform to the requirements of Military Specification MIL-P-63201.

3.2 Propellant grain. The inhibited propellant grain shall comply with the requirements specified on Drawing 9324786 and secondary drawings and with all requirements specified in applicable specifications.

3.3 Cracks, foreign material and cavitation.

3.3.1 Radiographic examination.

a. Cracks - There shall be no cracks as shown on the radiograph as dark lines in excess of .06 inches in length.

b. Foreign Material - There shall be no foreign material as shown by bright spots on the radiograph in excess of 0.06 inches in any shape or dimension.

c. Cavitation - There shall be no cavitation as shown by low density on the radiograph in excess of .03 square inches individually or a maximum combined area of .06 square inches in any 3 inches of length.

3.3.2 Visual examination.

a. Cracks - There shall be no cracks present within the propellant.

b. Foreign Material - There shall be no foreign material in excess of 0.06 inches in any shape or dimension.

c. Cavitation - There shall be no cavitation in excess of .03 square inches individually or a maximum combined area of .06 square inches in any 3 inches of length.

3.4 Inhibitors.

3.4.1 Separation/Delamination.

3.4.1.1 Inhibitors, excluding inhibitor cone. The propellant grain shall contain no individual separation between the inhibitor and the grain or between the inhibitor layers (delamination) which is greater than 0.01 square inch. The total combined area of propellant inhibitor separations/delaminations shall be no greater than 0.020 square inch. There shall be no propellant inhibitor separation/delamination within one inch of any split surface.

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3.4.1.2 Inhibitor cone. No separation shall be allowable at the cut edges of the inhibitor cone.

3.4.2 Exposed propellant. There shall be no exposed propellant at the bond lines of inhibitor segments.

3.5 Physical properties. Compression load and deformation.

3.5.1 Plus 140°F. The average of three (3) maximum compression readings from each lot shall be no less than 465 lbs., and the average of three (3) deformation readings from each lot shall be no less than 69 percent when tested as specified in 4.4.3.3.1.

3.5.2 Minus 40°F. The average of three (3) compression readings from each lot shall be no less than 9877 pounds and the average of three (3) deformation readings from each lot shall be no less than 11 percent when tested as specified in 4.4.3.3.2.

3.6 First article inspection. This specification contains technical provisions for both preproduction and initial production testing for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship.

3.7.1 Propellant, grain, assembly. The propellant grain assembly shall be free of dirt, grease, chips, gouges, solvent drippings, and other defects which would make the item unsuitable for its intended use.

3.7.2 Inhibitors. The inhibitors (wrapping tube and disc) shall be free from dirt, grease, chips, gouges, cracks, and other defects which would make the item unsuitable for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection
b. Quality conformance inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawing	Quantity
Inhibitor Cone, Aft Grain	9324785	10
Inhibitor End Disc	9324784	10
Propellant Billet (Wrapped)	9324783-2	10
Propellant Aft Grain	9324705	13
Grain, Assy, Aft	9324786	50
Propellant Billet (If Class 2 propellant is used)	9324782-2	10

4.3.1.1 Process changes. Any proposed change in the manufacturing process must be reviewed by the design agency, who at their discretion may waive or may increase the first article requirements for the Grain Assy, Aft. The increased first article will require a quantity of Sixty Four (64) Grain Assy, Aft. All other first article requirements will remain unchanged.

4.3.2 Inspections to be performed. The first article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and specifications and to the inspections of Table I. All inspections listed in Table I shall be classified Major unless otherwise specified in referenced paragraphs and specifications. See MIL-A-48078.

4.3.2.1 Ballistic tests.

4.3.2.1.1 Submission. Fifty (50) complete propellant grains packed in the form that will be used for shipping of the regular production quantity shall be submitted to a Government Installation as designated by the Contracting Officer for loading and ballistic testing. Unless otherwise directed by the Contracting Officer, the propellant grains shall not be submitted prior to the acceptance of the sample submitted to the Government for examination (see 4.3.1).

4.3.2.1.2 Tests to be performed. Proving Ground Tests shall be performed for information only using the procedures of 4.5.4.

4.3.2.2 Process changes. In the event the increased first article in 4.3.1.1 is required, the Sixty Four (64) Grain Assy, Aft shall be inspected as follows:

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Examination or Test	No. of Samples	Requir Para	Para. Ref/ Inspection Method
Examination for Defects and Tests	64	3.2	4.4.2.6
Radiographic Inspectiontion	64	3.2.1	4.5.1.2
12.2 Meter Drop Test (-40°F)	12	Info	TOP-4-2-601
12.2 Meter Drop Test (-60°F)	12	Info	TOP-4-2-601
Ballistic Test (-40°F)	10	3.2	4.5.4
Ballistic Test (-60°F)	10	3.2	4.5.4
Ballistic Test (+70°F)	10	3.2	4.5.4
Ballistic Test (+145°F)	10	3.2	4.5.4

The ballistic and drop test samples shall be loaded into inert warhead/live rocket motor M549A1 projectiles. After the drop test is completed, the sample projectiles shall be radiographed in two orientations 90° apart. The radiographs as well as other test results will be submitted to ARRADCOM, DRDAR-QAR-Q to determine if a ballistic test is required.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Propellant Grain, Aft				See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inhibitor, End Disc (Dwg. 9324784-2) Examination for Defects and Tests	10	-	3.2	4.4.2.1
	Inhibitor, Cone Aft Grain (Dwg. 9324785) Examination for Defects and Tests	10	-	3.2	4.4.2.2
	Propellant Billet (Dwg. 9324782-2) Examination for Defects and Tests	10	-	3.2	4.4.2.3
	Propellant Billet (Wrapped) (Dwg. 9324783-2) Examination for Defects and Tests	10	-	3.2	4.4.2.4

NOTES:

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TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	CONTINUATION SHEET				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Propellant Aft Grain (Dwg. 9324705)				
	Examination for Defects and Tests	10	-	3.2	4.4.2.5
	Physical Properties	3	-	3.5	4.5.5
	Propellant Grain, Assy Aft (Dwg. 9324786)				
	Examination for Defects and Tests	50	-	3.2	4.4.2.6
	Ballistic Test	50	-	3.2	4.5.4
	Radiographic Inspection	50	-	3.3.1	4.5.1.2

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

In addition, each lot of propellant grains shall contain not more than one lot of propellant. The lot interfix number shall be changed when a new lot of propellant is introduced into the rocket grain extruding process. Additionally, a grain lot shall be no less than one week or more than one month scheduled production of rocket grains.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9324784
4.4.2.1	Inhibitor, End Disc				NEXT HIGHER ASSEMBLY 9324705
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Thickness	a	0.40%	3.2	Gage
102	Excessive warpage	a	0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship	a	1.00%	3.7.2	Visual

NOTES:
a. See 4.4.3.1.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63203 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9324785
4.4.2.2	Inhibitor, Cone Aft Grain				NEXT HIGHER ASSEMBLY 9324786
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Overall length (min.)	a	0.40%	3.2	Gage
102	Angle of cone	a	0.40%	3.2	Gage
103	Thickness	a	0.40%	3.2	Gage
104	Inside diameter of small end	a	0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship	a	1.00%	3.7.2	Visual

NOTES:
a. See 4.4.3.1

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9324782-2
4.4.2.3	Propellant, Billet (after OD Machining)				NEXT HIGHER ASSEMBLY 9324783-2
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Outside diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanip		1.00%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63203 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9324783-2
4.4.2.4	Wrapped Propellant Billet				NEXT HIGHER ASSEMBLY 9324705
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None Defined.				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inhibitor improperly wrapped		0.40%	3.2	Visual
103	Radiographic inspection		100%	3.3.1	4.5.1
Minor					
201	Evidence of poor workmanship		1.00%	3.7.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63203 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Propellant Aft Grain				9324705
					NEXT HIGHER ASSEMBLY 9324786
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Total length (max.)		0.40%	3.2	Gage
102	Location of small basic diameter on taper		0.40%	3.2	Gage
103	Location of large basic diameter on taper		0.40%	3.2	Gage
Minor					
201	Diameter of inside diameter chamfer		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.7.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.6	Propellant Grain Assy, Aft				9324786
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Excessive cracks, foreign material, or cavitation		100%	3.3.2	Visual
102	Exposed propellant at intersections of inhibitor surfaces		100%	3.4.2	Visual
103	Inhibitor separation/delamination		0.40%	3.4.1.1	Visual
104	Finished weight of grain		0.40%	3.2	Balance
105	Spacers missing or improper		0.40%	3.2	Visual
106	Poor joint or improper blending of top and bottom and side inhibitor		0.40%	3.4.1	Visual
107	Surface condition of saw cut surface		0.40%	3.2	Visual
108	Width of sawcut	a	-	3.2	4.5.2

NOTES:
a. The first item at the beginning of each 8 hours production shall be used for this inspection.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.6	CONTINUATION SHEET				9324786
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
109	True position of sawcuts	a	-	3.2	4.5.3
110	Cone separation		0.40%	3.4.1.2	Visual
111	True position of inside diameter to outside diameter		0.40%	3.2	Gage
112	Excessive adhesive		0.40%	3.2	Visual
Minor					
201	Inside diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.7	Visual

NOTES:
a. The first item at the beginning of each 8 hours production shall be used for this inspection.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Packing, Aft Grain				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined.				
Major					
101	Drum split or damaged		0.40%	5.1	Visual
102	Tape missing or improperly applied		0.40%	5.1	Visual
103	Protective bag damages		0.40%	5.1	Visual
Minor					
201	Marking incomplete or incorrect		0.65%	5.2	Visual
202	Evidence of poor workmanship		1.0%	3.7.4	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Inhibitor cone/end disc inspections. The following sampling sequence may be used in lieu of sampling per the AQL listed in the respective paragraphs.

4.4.3.1.1 The first consecutively produced pieces from each set of dies shall be inspected for dimensional characteristics as shown on the applicable drawing and applicable paragraph in this specification until twenty (20) consecutively produced acceptable items are found. If five hundred (500) pieces are produced without obtaining 20 consecutive acceptable items, production shall be stopped, and the process corrected prior to producing any additional components.

4.4.3.1.2 After meeting the requirements of 4.4.3.1.1, five (5) pieces from each set of dies shall be selected at intervals of 1,000 pieces from each set of dies or mold cavities and inspected for dimensional characteristics as specified on the applicable drawing and applicable paragraph in this specification. The occurrence of any defect on any sample piece shall be cause for taking corrective action, for rejecting all pieces produced since the last acceptable sample, and returning to sampling per 4.4.3.1.1.

4.4.3.2 Radiographic inspection.

4.4.3.2.1 Billet inspection - Major defect. The propellant billet shall be subjected to 100% radiographic inspection. This test shall be conducted in accordance with 4.5.1.1.

4.4.3.2.2 Grain inspection - Major defect. One sample of a completed grain selected per 8 hours production shall be radiographed. In the event the sample grain fails to meet the requirements of 3.3.1, all grains produced after the previous radiographic sample shall be radiographed and any additional grains failing to meet the requirements of 3.3.1 shall be removed from the lot. This test shall be conducted in accordance with 4.5.1.2.

4.4.3.3 Physical properties.

4.4.3.3.1 Plus 140°F - Major defect. Three (3) samples will be randomly selected from the lot. If possible, no two samples will be taken from the same billet. These samples shall be tested for maximum compression load and deformation as specified in 4.5.5. The lot shall be rejected if the average of the three specimens from any casting fails to meet the requirements of 3.5.1.

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4.4.3.3.2 Minus 40°F - Major defect. Three (3) samples will be randomly selected for the lot. If possible, no two samples will be taken from the same billet. These samples shall be tested for compression load and deformation as specified in 4.5.5. The lot shall be rejected if the average of the three specimens from any casting fails to meet the requirements of 3.5.2.

4.4.3.4 Width of cut - Major defect. The width of cut taken in splitting the grain shall not exceed the drawing requirement. For verification, the width of cut shall be inspected at the beginning of each 8 hours of production for compliance with 3.2.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and Section 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Radiographic inspection.

4.5.1.1 Billet inspection. The propellant billet shall be radiographed with the beam parallel to the longitudinal axis of the billet. Observations shall be made for cracks, voids, fissures and foreign material in accordance with MIL-STD-453.
A penetrameter of sample test block showing the types of defects for rejection shall be designed to assure a 2 percent minimum sensitivity.

4.5.1.1.1 Fluoroscope. In lieu of x-ray equipment, fluoroscopic equipment can be utilized for radiographic inspection of the propellant billet. The fluoroscopic equipment shall have the capability to permanently record the fluoroscopic images, record the comments of the inspector, photograph selected defects, and measure defect sizes. An Image Quality Indicator should be in the field of view at all times to simulate cavity and crack type defects.

4.5.1.2 Grain inspection. The split finished machined propellant grain shall be radiographed with the beam parallel to the longitudinal axis of the grain. Observations shall be made for cracks, voids, fissures and foreign material in accordance

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with 3.3.1. Radiographic inspection procedures shall be in accordance with MIL-STD-453. A penetrameter or sample test block showing the types of defects for rejection shall be designed to assure a 2 percent minimum sensitivity.

4.5.2 Width of saw cut. The width of the sawcut shall be measured by making a partial cut through the sample of paragraph 4.4.2.6 and then removing the saw blade and measuring the width of cut.

4.5.3 True position of saw cut. The sample used in 4.5.2 shall be gaged after the partial cut for true position of the saw cuts.

4.5.4 Ballistic test. The 50 grain assemblies manufactured during the first article shall be inspected per Table I and then shipped to a specified Government loading facility. These grains shall be loaded into a M549 (Inert Warhead/Live Motor Body) test vehicle and fired at a Government Test Facility. Test procedures to be used are those contained in the M549 Projectile Loading, Assembly and Packing specification (MIL-P-50578). Failure of the propelling grains to perform according to the criteria of MIL-P-50578 shall be cause for rejection of the first article sample.

4.5.5 Physical properties. Each specimen and specimen support shall conform to the requirements of Figure 1. The compression testing shall be conducted in a test fixture equipped with a temperature conditioning box as an integral part of the fixture. The specimen shall be inserted between the fixture posts with steel cylinders the same diameter as the specimen and placed above and below the specimen forming one continuous concentric cylinder. The specimen shall be conditioned at the required temperature plus or minus 5°F for three hours minimum and shall be tested at a crosshead speed of 2 inches per minute.

5. Packaging

5.1 Specific Requirements

5.1.1 Level A. Twenty-four (24) aft grains shall be packaged in an RSC Style, B Flute, SW, Domestic Class, 175 minimum fiber board in accordance with Specification PPP-B-636.
The size of the fiber box is to be as follows: 20 x 15 x 6 (+ 1/8 tol) inches inside and 14 3/4 x 4 3/4 inches between flaps (top and bottom). The grains shall be packaged in two layers of twelve (12) per layer. Each grain shall be inserted into a spiral or convolute wound chipboard tube having the following dimensions:

Inside diameter	-	4.650 + .015 inches
Outside diameter	-	4.975 - .030 inches
Length	-	2.860 - .020 inches

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A separator, 19 1/2 x 14 1/2 inches, of the same material as the fiber box shall be placed between layers. Additional separators shall be placed on the top, on the sides (19 1/2 x 5 1/2) inches and on the ends (14 1/2 x 5 1/2) inches as required for tight packing. The maximum moisture content of the paper products shall be 9% prior to packout. Close the fiber box and seal it with a center sealing strip using Class I, Type III, any color, specification PPP-T-60 tape. Blunt the corners of the fiber box and place in a Type I, Class E, Style I, specification MIL-B-117 Barrier bag. The size of the barrier bag shall be approximately 22 1/2 inches openface by 29 3/4 inches deep. Exhaust excess air from the bag and seal in accordance with the manufacturer's instructions.

5.1.2 Level B. See 5.2.2.

5.2 Packing

5.2.1 Level A. Overpack a bagged fiber box in a Grade A, Type II, Class I, specification MIL-B-2427 wood box. Inside dimensions of the box shall be 21 x 15 1/2 x 6 3/4 (+ 1/8 tol) inches. One filler shall be used on the inside, the top and at each end. Additional fillers shall be used as required for tight packing. The fillers shall comply with specification MIL-F-50449. The box cover shall be positioned and the box shall be strapped with 2 straps in accordance with specification MIL-B-2427.

5.2.2 Level B - (Not for Overseas shipments). Twelve (12) aft grains shall be packed in a fiber drum meeting the requirements of DOT specification 21C of the Code of Federal Regulations 49 CFR Parts 100 to 199. The drum wall shall be lined with a kraft backed commercial grade single-faced corregated fiber board using as many layers as required for a tight pack. The drum size shall be approximately 11 5/8 x 11 5/8 x 9 inches high. A barrier bag meeting requirements of MIL-B-117, Type I, Class E, Style I shall be placed inside the lined drum and 12 propellant grains shall be packed in 3 layers of 4 per layer. The barrier bag size shall be approximately 24 x 22 1/4 inches. Each grain shall be inserted into a spiral or convolute wound chipboard tube having the following dimensions:

Inside Diameter	-	4.650 + .015 inches
Outside Diameter	-	4.975 - .030 inches
Length	-	2.860 - .020 inches

The layers of the chipboard tube shall be placed in the fiber drum with the marked end up. A separator shall be placed between layers and on top of final layer. The separator shall meet the requirements of PPP-F-320, Type CF, Class Domestic, Grade 200, B or C Flute material or equivalent. The bag shall be folded and sealed

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with tape meeting requirements of PPP-T-60 or equivalent. Filler material constructed of the same material as the separators shall be placed on top of the contents as required to provide a tight pack. The fiber drum cover shall be sealed with 2-inch, (minimum) wide tape meeting the requirements of PPP-T-76 or equivalent. The contractor shall certify that all packing material inside the barrier bag shall not exceed 9 percent maximum moisture content.

5.3 Marking.

5.3.2 Level A. The fiber box shall be marked in black ink with 1/4 inch minimum height lettering. The lettering shall include the following information:

Grain Propelling Aft F/RAP
155MM, M549A1
Date Packed (Mo./Yr.)
Lot No. (Insert No.)

Marking on the bag shall be in water resistant ink. The wood box is to be marked in accordance with Drawing 8796522. DOT to be:

Propellant Explosives (Solid) Class B
Grain Propelling Aft F/RAP
155MM M549A1
Date Packed (Mo./Yr.)
Lot No. (Insert No.)

5.3.2 Level B. The fiber drum shall be marked as follows:

DOT marking on top - "PROPELLANT EXPLOSIVES (SOLID) CLASS B

On Side Wall - "12 GRAIN, PROPELLANT, AFT F/RAP 155MM, M549A1
Date Packed (Mo./Yr.)
Lot No. (Insert No.)
WT. (Insert Wt.) CU (Insert Volume)

Marking shall be with black waterproof ink and lettering shall be 1/4 inch minimum height.

6. NOTES

6.1 Intended Use. The components covered by this specification are intended for use of the 155MM, HE Rocket Assisted Projectile, M549A1.

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6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection procedures and/or equipment designs for approval - See MIL-A-48078. Submit equipment designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, NJ 07801.

6.4 Inspection of true position of saw cut (see 4.4.2.6). In lieu of performing this specification, the Contractor may verify the true position by virtue of tooling by submitting appropriate designs for approval to control the true position.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:
Army - AR

Preparing activity:
Army - AR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-63203	PROJECTILE, 155MM, HE, RA, M549A1 AFT (M37 EXTRUDED) PROPELLANT GRAIN FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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